CLASS 9 · CBSE · GANITA MANJARI PART 2 · FORMULA BANK

Chapter 10: How Quantities Combine: Understanding Data

Every formula and result of the chapter on one sheet

1	Sum of a group	Average $\times$ number of values
2	Combined average of two groups	$\frac{an + bm}{n + m}$
3	Combined average of three groups	$\frac{ap + bq + cr}{p + q + r}$
4	Groups of equal size	Combined average $=$ average of the averages
5	Concentration	$\frac{\text{quantity of the substance}}{\text{quantity of the mixture}}$
6	Concentration of a mixture	$\frac{w_1x_1 + w_2x_2}{w_1 + w_2}$, with quantities w and concentrations x
7	Weighted mean	$\frac{w_1x_1 + w_2x_2 + \ldots + w_nx_n}{w_1 + w_2 + \ldots + w_n}$
8	After adding a value v	$\frac{an + v}{n + 1}$
9	After removing a value v	$\frac{an - v}{n - 1}$
10	Percentage share	$\frac{\text{part}}{\text{total}} \times 100$
11	Stacked bar chart	Shows the total and its parts
12	100% stacked bar chart	Shows shares only; every bar is 100%